

SIMATS ENGINEERING
ASSESSMENT TOOL 1 - Low (Pseudo-code Writing)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO1: *Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)*

QUESTIONS: 10

Total Marks: 100

ANNEXURE A

PSEUDO-CODE WRITING

Code of Conduct:

I N.Adi Narayana(192525346)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
N.Adi

① student management system

class student

& Private name, roll no, marks1, marks2, Average
method accept Data()

&
Read name, roll no, marks1, marks2

&
Method calculate Average()

&
 $Average = (marks1 + marks2) / 2$

&
Method display Result()

&
Display name, roll no, average

If average ≥ 50

Display "Pass"

else

Display "fail"

&

&

②

Library

Book

Management

class Book

&

Private title, author, ISBN, is Available

method issueBook()

&

is available = "false"

&

method returnBook()

&

is Available = true

&

method displayBook()

Display title, author ISBN, available

}

Main()

{
Create Books

Create Books

Create Books

Display all Books

}

}

Employee salary calculation

class Employee

{
private EmpId, Name, Basic Salary

{
Gross Salary = basicPay + HRA + DA

}

Method CalculateNetSalary()

{
PF = basicPay * 5%

Tax = basicPay * 10%

NetSalary = grossSalary - PF - Tax

}

Method displaySlip()

{
Display EmpId

Display name

Display designation

Display netSalary

}

Main()

{
create employee object

displaySlip()

④ Shape Area Calculator

class Shape

&
Method calculateArea()

↳
class Circle extends Shape

&
~~Area = 3.14 x r x r~~

↳
class Rectangle extends Shape

&
~~Area = 0.5 x b x h~~

↳
Main()

&
calc calculateArea()
display

↳

⑤

Vehicle Registration System

class Vehicle

&
Method displayDetails()

&
Display Vehicle Details

↳

↳
class TwoWheeler extends Vehicle

&

↳
class FourWheeler extends Vehicle

&

↳
Main()

↳

file

⑥ Bank Account Hierarchy

class BankAccount

⊗ accNo, holder, balance

deposit()

withdraw()

↳ class Saving Account extends BankAccount

⊗ interestRate

calculateInterest()

↳

class Current Account extends BankAccount

⊗

overdraftLimit

withdraw()

↳

main()

⊗ create Saving & Current Account

↳

⑦

Hospital Patient Record System

class Patient

⊗ Protected patientId, name

Private diagnosis, prescription

method generateBill()

⊗

↳

↳ class Inpatient extends Patient

⊗

wardNo

admission Date

daily charges

method generateBill()

Bill = days x daily charges

Display bill

↳

8

E-Commerce Product Catalog

class Product

Product ID, name, price

Apply Discount()

class Electronics extends Product

warranty

class Clothing extends Product

size

Main()

create Electronics object

create Grocery object

}

9

University staff Hierarchy

class staff

staff ID, name

calculate Allowance()

class Teachingstaff

Subject

Main()

create professor object

create Lecturer object

create nonTeachingstaff object

}

10

Smart Home Device Controller

class SmartDevice

{
device name
executeCommand()
}

{
class smartlight extends SmartDevice

{
class smartthermostat extends SmartDevice

{
class smartlock extends SmartDevice

{
main()

{
create smartlight object
create smartlock object
}

